

# Krishnaa Vadivel

Scientific Machine Learning | Computational Materials

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Portfolio: <https://krishnaa423.github.io/personal-website/> | GitHub: <https://github.com/krishnaa423>

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## Summary

Scientific-ML researcher with a physics-first background in equivariant graph networks, automatic differentiation, first-principles modeling, and GPU/HPC implementation.

## Technical Skills

**ML:** JAX, PyTorch, PyTorch Geometric, e3nn, equivariant GNNs, automatic differentiation, NumPy, SciPy

**Scientific compute:** Python, C++, MPI, OpenMP, CUDA (cuBLAS, cuTENSOR), ROCm (rocBLAS, rocFFT), PETSc, SLEPc, DOLFINx

**Platforms:** Frontier, Perlmutter, Frontera, Docker, Kubernetes, Linux

## Experience

### PhD Researcher

*Yale University, Electrical Engineering*

2021–present

- Build DeePH-style computational graphs for LiF and double-perovskite systems; train equivariant models in PyTorch Geometric and use automatic differentiation to obtain force and optical-property quantities.
- Connect ML models to first-principles exciton and many-body theory, ensuring datasets, losses, and evaluations map to physical observables.
- Build and debug distributed GPU workflows in Python/C++ using MPI, OpenMP, CUDA/ROCm, PETSc, and SLEPc on national HPC systems.
- Prototype retrieval-driven scientific assistants that extract structured simulation inputs from papers and technical documentation.

### Software and Embedded Systems Engineer

*Probe Technologies / Weatherford Wireline Services, Houston, TX*

2019–2021

- Built CUDA-enabled waveform-processing and visualization tools, adding production experience with data pipelines and performance constraints.
- Implemented SQL-backed ASP.NET services and C# tools for remote engineering-test workflows.

## Selected Projects

**EGNN QM9 force model** — Trained an equivariant GNN and differentiated predicted molecular energies to obtain forces.

**Materials optical-property models** — Applied graph networks and automatic differentiation to accelerate costly calculation stages; roughly 100x speedup in parts of the workflow.

**Scientific retrieval pipeline** — Structured software documentation and paper content into starter artifacts for computational-science workflows.

## Education

### PhD, Electrical Engineering

*Yale University, Computational materials and scientific ML*

2021–present

### MPhil and MS, Electrical Engineering

*Yale University*

2023

### MEng and BS, Electrical and Computer Engineering

*Cornell University,*

2013–2018